

e-waste from computers will jump by 200 to 400 per cent from 2007 levels, in South Africa and China. For India, this figure is predicted to be 500 per cent. As per a research, by 2030, the obsolete PCs from developing countries will reach 400-700 million units, which is far more than from the developed countries at 200-300 million units. Rapid product innovation and upgradation with decreased price will lead to these facts and figures.

While e-waste is a big issue, unnecessary consumption of electricity in IT is another priority issue. Research says that the manufacturing of a computer requires about 1818 kilowatt per hour of electricity. A PC consumes about 200-300 watts in 'on' state. Researches show that numerous PCs are left switched-on and idle continuously, resulting in huge electricity wastage everyday. As of today, electricity is generated mainly by burning fossil fuels, which emit pollutants such as carbon dioxide and sulphur etc, in the air, which are responsible for global warming. Thus green computing has to have a two-fold approach—reducing e-waste and increasing efficient use of electricity and/or new clean energy sources.

Reduction of waste and recycling of the used and obsolete computers and peripherals are essential for green computing. Several approaches are attempted. New material research and application of bio and nanotechnology are likely to save the situation. In Europe, Restriction of Hazardous Substance (RoHS) programme that makes use of lead-free computing has been taken up. Similarly, Waste Electrical and Electronic Equipment (WEEE) is designed to reduce the effects of hazardous waste.

Efficient use of energy and power savings may have several options:

1. Adoption of energy star standard. Computing systems must meet certain consumption requirements when idle and when fully powered. For instance, their power supplies must be able to convert 80 per cent of incoming electricity for use by the

PC. For computer a specification of energy star 4.0 became effective from July 20, 2007 with 80 PLUS Bronze label for power supply unit. 80 PLUS label guarantees more than 80 per cent energy efficiency at 20 per cent, 50 per cent and 100 per cent of rated load, and a power factor of 0.9 or greater at 100 per cent load. It is reported that IBM adopted 'lean six sigma' principles of efficiency.

By the process, in 1990 IBM saved around 4.6 billion kilowatts of electricity and prevented about 3 million metric tonnes of carbon dioxide emission.

The Indian University School of Education, is reported to have, in a pilot project, programmed the desktop computers to go to sleep after several hours of constant idleness. Energy consumption decreased by 48.3 per cent for the desktop computers in the said study. Based on the findings, it was estimated that instituting similar computing measures across the entire university campus could save more than \$500,000 per year in energy-costs and reduce emissions of carbon dioxide by more than 15,000 tonnes every year.

Oxford University is doing a pioneering research on energy-saving low-carbon green computing. Researcher Daniel Curtis, of Oxford University's Environmental Change Institute, reportedly said, "No-one sits at their computer for 168 hours a week. When a computer is switched on, its power demand remains pretty much constant, regardless of whether its user is surfing the net, word-processing, or at home in bed. We are developing a system that will mean that computers only need to be switched on when actually in use."

2. Judicious choice of system. A laptop consumes five-times less energy than a desktop PC. Therefore consumers must be guided to use more laptops than desktops for realisation of green computing culture.

3. Virtualisation. It is a technique



of running more than one system with single set of physical hardware. Virtualisation allows minimum power consumption and maximum cooling effect. Wherever and whenever possible, virtualisation may be made a habit. Examples are Useful and Thin Network clients. Useful develops software for Linux-based remotely-managed virtual desktops that enables up to ten users to share one computer at a time.

4. Telecommuting. It is an experience of working from home. It is virtual working that reduces overhead costs like light and electricity in offices and makes a habit of green IT uses.

5. Use of new and renewable energy resources. Solar-based computing tools and devices, for example, require a research culture for green computing.

6. Use of power saving system. Use may be made of power saving options of Windows operating system and power saving processor. For example, the Intel Core 2 Duo desktop processor is up to 40 per cent faster and energy-efficient than its single-core predecessor.

Choice of a proper graphics card is a key factor for power saving computing. If advanced 3D games are not required, integrated graphics for the lowest power consumption may be adopted. ●

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